INTERIM FEDERAL SPECIFICATION

MOPS, DUSTING (RENTAL SERVICE)

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Washington, D. C. 20405, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for treated cotton dust mops, covers, and accessories, primarily intended for dusting floors, to be furnished to the Government by commercial services on a rental basis. The items furnished shall be laundered, treated, and maintained in good serviceable condition by the supplier.

1.2 Classification.

1.2.1 Items, types, classes, and sizes. Unless otherwise specified in the invitation for bid, the items to be supplied shall be of the following types, classes, and sizes (see 6.2). Sizes not listed herein may be specified as required at the option of the procuring officer.

Item I. Dust mopping tool (handle and frame).

Size 12 - 12 inches	Size 24 - 24 inches
Size 14 - 14 inches	Size 30 - 30 inches
Size 18 - 18 inches	Size 36 - 36 inches
Size 20 - 20 inches	Size 42 - 42 inches
Size 22 - 22 inches	Size 60 - 60 inches

Item II. Mophead, cotton, treated (to be used with item I).

Sizes for this item shall be the same as for item I (see 3.1.2.2).

Item III. Dust sweeping tool (handle and brushhead).

Size 12 - 12 inches
Size 20 - 20 inches
Size 42 - 42 inches

Item IV. Cover, cotton, treated (to be used with item III).

Type 1 - Flat.
Class 1 - Cloth, cotton, osnaburg
Size 12 - 18 by 30 inches (greige size)
Size 20 - 36 by 36 inches (greige size)
Size 42 - 36 by 55 or 36 by 60 inches (greige size)

Type 2 - Tubular.
Class 1 - Cloth, cotton, osnaburg
Sizes for type 2 shall be the same as for type 1.

2. APPLICABLE DOCUMENTS

2.1 The following specification and standards of the issues in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. The mop assemblies to be supplied under this specification shall be available in the types, classes, and sizes as specified in 1.2.1. All components of the mop assemblies shall be in good operating condition when received by the Government and so maintained by the supplier. Each mophead and brushhead cover shall be properly treated with a sufficient quantity of a dust adhesive compound to effectively pick up and retain dust, soil, and lint to such an extent that when the fingers of the hand are swept once over the freshly cleaned area, there shall be no evidence of dirt on the finger tips. The mopheads and covers shall be clean in appearance having no holes, tears, or ragged edges, except that the brushhead covers may be repaired within a radius of two inches of the center hole. All metal parts shall be of corrosion resistant metal or metal that has been coated or plated to resist corrosion.

3.1.1 Item I. Dust mopping tool. The mopping tool shall consist of a wood handle (see 3.1.5) and a metal mophead holder or frame. The tool shall be designed to permit ready insertion into the mophead pockets or sleeve and shall be attached to the handle by means of a universal joint, or functional equivalent, which allows the frame to swivel 360 degrees. The handle shall lower at any operating point on the tool to such a position that the end of the handle shall be not more than 1-3/4 inches from the floor. The holder or frame shall be 5 inches wide, $\pm$ 1/2 inch, measured at the widest portion, and the lengths shall be as specified in 1.2.1, $\pm$ 1/2 inch. If designed with fill pieces, they shall be made of rigid durable plastic material and securely attached to the frame. A coupling device may be furnished which will permit the joining of two frames when required for the cleaning of extra large areas. If so provided, the coupling device shall be capable of installation without the aid of tools and shall not interfere with individual use of handle or handles on frames so joined.

3.1.2 Item II. Mophead, cotton, treated. The mophead to be used with the dust mopping tool (item I) shall be of sleeve or pocket design. The mophead boot shall be made of cotton duck of suitable weight and of the best quality for the purpose intended. The yarn used on the mophead shall be of a suitable grade and staple to meet the requirements of this specification.

3.1.2.1 Construction. All cotton duck components, top and bottom, shall be securely sewn together and bound or overedged around the edges. The pockets or sleeve shall be designed to allow for ready insertion of the holder or frame and shall be bound or otherwise reinforced at the openings. At the option of the supplier, the yarn may be sewn or tufted to the mophead boot. If the yarn is sewn on, there shall be not less than three rows, not including the row of yarn around the outer edge, securely sewn on the bottom of the boot. If tufted to the boot, there shall be not less than fifteen rows of yarn, excluding the outer or fringe row, securely tufted on the bottom of the boot. There shall be at least one additional row of yarn (fringe) around the periphery of the mophead. The inner rows of sewn or tufted yarn shall be not less than 1 inch nor more than 1-3/4 inches in length and the fringe row of yarn shall be not less than 2-1/4 inches in length, measured from point of penetration into the boot to the yarn tip.

3.1.2.2 Dimensional requirements. Mopheads shall be available in the sizes corresponding to the holder or frame (1.2.1) and shall fit the applicable size frame in such a manner as to preclude rolling in service. Sizes shall be measured by boot length excluding the length of yarn which extends beyond the boot. Each mophead shall be clearly marked to identify it with the applicable size frame. Unless otherwise specified (see 6.2), the spread size (measurement from yarn tip to yarn tip) may be indicated at the option of the supplier.

3.1.3 Item III. Dust sweeping tool. The sweeping tool shall consist of a replaceable wood handle (see 3.1.5) and a brush block or frame (brushhead) with a universal type swivel. The brush block or frame shall be of close-grained hardwood, smoothly finished and uniformly coated with varnish, lacquer, or other suitable material in accordance with the manufacturer's standard commercial practice. The block or frame may be solid or laminated. If laminated, the wood shall be bonded with moisture-resistant adhesives. The brushhead shall be 5 inches, $\pm$ 1/2 inch, wide and in the lengths specified in 1.2.1, $\pm$ 1/2 inch. The brushhead face shall contain not less than two rows of hair, bristles, or filament of such quality and stiffness as to retain its original shape during normal usage. The filler material shall be not less than 1-1/4 inches nor more than 2 inches in length and shall be placed around the perimeter of the block in such a manner as to be capable of holding the cloth cover firmly to flat surfaces. Additional rows of filler material, if any, shall be not less than 1/2 inch in length. The weight of the dust sweeping tool assemblies (handle and brushhead) shall be as follows:

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Weight requirements

Size	Weight in pounds	
	Minimum	Maximum
12	2-1/4	2-1/2
20	2-1/2	3
42	3-1/2	4

3.1.3.1 Universal joint. The brushhead shall be attached to the handle by means of a universal joint which shall permit the brushhead to rotate 360 degrees when tested as specified in 4.4.2. The universal joint shall have sufficient friction to prevent the brushhead from swiveling freely when held off the floor and yet not interfere with its rotation when on the floor.

3.1.4 Item IV. Cover, cotton, treated. The covers to be used with the dust sweeping tool (item III) shall be of two types, flat and tubular, and shall be made from one piece of material. The material shall be cotton osnaburg of a weight and quality best suited for the purpose intended.

3.1.4.1 Type 1. Cover, flat style. The finished dimensions of the covers shall meet the requirements for the respective size tool as indicated below. The edges shall be selvage or overedged. There shall be an overedged or otherwise reinforced center hole not less than 1-1/2 inches nor more than 2-1/2 inches in diameter.

Dimensional requirements of covers

Sweeping Tool Size	Maximum (Inches)		Minimum (Inches)	
	Width	Length	Width	Length
12	18	30	15	22-1/2
20	36	36	27	30
42	36	55	30	42
	or			
	36	60	30	45

3.1.4.2 Type 2. Cover, tubular style. The type 2 cover shall be identical to the type 1, except that two edges shall be securely sewn together to form the sleeve-like or tubular style and there shall be two overedged or otherwise reinforced handle holes so that all portions of the cover may be used in the sweeping process. The dimensional requirements shall be the same as specified in 3.1.4.1, except that in order to determine compliance, the stitching joining the ends shall be removed before the cover is measured.

3.1.5 Handle. The handles to be used with the dusting tool (item I) and the sweeping tool (item III) shall be of hardwood, smoothly finished and coated with varnish, lacquer, or other suitable clear organic material in accordance with the manufacturer's standard commercial practice. The diameter of the handles shall be not less than 7/8 inch nor more than 1-1/8 inches and the length shall be not less than 59 inches nor more than 64 inches. There shall be a means for secure attachment of the handle to the dusting tool with which it is to be used. The handle shall be free from imperfections such as splinters, decay, or defects which may be injurious to the user or affect serviceability.

3.2 Impregnation. The mopheads and brushhead covers shall be impregnated with a dust absorbent compound conforming to the requirements of 3.3 and applied by commercially acceptable methods (see 3.4). The treating compound shall be uniformly dispersed throughout the mopheads and brushhead covers and shall be present to the extent of 20.0 percent, minimum, and 35.0 percent, maximum, of the untreated dry weight of the cotton mophead, and 8.0 percent, minimum, and 23.0 percent, maximum, of the untreated dry weight of the brushhead cover when tested in accordance with 4.4.3. A certificate of compliance for this requirement will be acceptable (see 4.2.1).

3.2.1 The treated mopheads and brushhead covers shall not be tacky or greasy to the touch, or leave visible evidence of oil residues on cleaned surfaces to the extent of not showing a deposit of oil on a flat nonabsorbent surface when tested in accordance with 4.4.4.

3.3 Treating compound. The dust control compound for treating the mopheads and brushhead covers shall consist of a paraffinic or naphthalenic base petroleum product, germicides, and a suitable aromatic when required. The compound shall contain not more than 5.0 percent water and no animal, or vegetable oil, or any ingredient that will permit or cause it to heat or ignite spontaneously, or cause injury to the skin, and shall have an Underwriters Laboratories Fire Hazard Rating between 0 and 20. The compound shall contain sufficient bacteriological additives to inhibit the growth and spread of environmental bacteria, such as Staphylococcus aureus, and reduce contamination on

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the areas cleaned. When tested for this requirement the Zone of Inhibition shall be at not less than 1 mm. A laboratory report shall be submitted certifying compliance with this requirement.

3.3.1 Laboratory report approval. Unless otherwise specified (see 6.2), prior to or at time of submission of bid, the bidder shall submit to the contracting officer a certified copy of a recent laboratory report which indicates that the items proposed for delivery have been tested and found to comply with the bacteriostatic action requirements of this specification. Either of the following types of reports will be satisfactory:

(a) An independent or commercial laboratory report.
(b) The prospective supplier's own laboratory report.

3.4 Method of treating. The compound shall be applied to the mopheads and brushhead covers by machine, such as by spraying or absorption in washwheels, in a manner consistent with the best commercial practices.

3.5 Dirt pick-up efficiency. The dirt pick-up efficiency of the mophead shall average not less than 40 percent and the brushhead cover not less than 20 percent based on the weight of the treated item before soiling, when tested as specified in 4.4.5.

3.6 Stability. The treated mopheads and covers shall not lose their effectiveness or have a tendency to sour or mildew under normal storage conditions.

3.7 Color. Mopheads and brushhead covers shall be unbleached or a dyed color at the supplier's option. If dyed, the dye used shall in no way interfere with the dust absorbent treatment.

3.8 Odor. The treated mopheads and brushhead covers shall be free from objectionable odors.

3.9 Identification marking. Each item shall be marked to clearly identify the supplier by name or trademark of such known character that the source of supply can be readily determined.

3.10 Directions for use. Unless otherwise specified(see 6.2), the supplier shall provide instructions for the operation and use of the tools. At the option of the procuring activity, these instructions may be in printed from or by means of suppliers' demonstration.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may use his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Component and material inspection. In accordance with 4.1, the supplier is responsible for insuring that components and materials used were manufactured, examined, and tested in accordance with this specification or referenced subsidiary specifications, unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.3 Sampling and inspection for lot acceptance. Sampling and inspection shall be performed in accordance with MIL-STD-105. For sampling purposes, a lot shall consist of all items of the same type, class, and size submitted for delivery at one time.

4.3.1 Examination of the end item. Each sample unit shall be examined for the applicable visual defects listed in table I. Inspection level S-2 shall apply and the acceptable quality level (AQL) shall be 4.0 defects per hundred units.

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TABLE I. Visual defects in end item.

Examine	Defects
Item I - Mopping tool	Evidence of rust on metal parts. Design not as specified. Universal joint missing or does not permit 360 degree swivel action. Handle does not lower to required distance from floor. Any sharp burr on surface which might cause personal injury.
Item II - Mophead	Design not as specified. Boot material and cotton yarn not suitable for purpose intended. Edges not bound or overedged. Number of rows and lengths of yarn less than specified. Objectionable odor. Spotted, soiled, or stained.
Item III - Sweeping tool	Evidence of rust on metal parts. Block not smoothly finished and coated. Number of rows and lengths of bristle not as specified. Universal swivel joint missing or does not permit 360 degree swivel action. Dimensions and weight not as specified.
Item IV - Brushhead covers Types 1 and 2	Not fabricated from one piece of material. Dimensions not as specified. Edges not selvage or overedged. Handle hole or holes not reinforced as specified. Not free of tears or ragged edges. Objectionable odor. Spotted, soiled or stained.
Item V - Handles (To be used with items I and III)	Not smoothly finished or coated as specified. Evidence of rust on metal parts. Not equipped with adequate means for secure attachment to applicable tool. Dimensions not as specified. Not free from splinters, decay, or burrs which might cause personal injury.
Identification marking	Omitted, illegible, incomplete.

4.3.2 Sampling for testing (items II, III, and IV). Unless otherwise specified, three sample units each of item II (mopheads), item III (sweeping tools), and item IV (covers) shall be selected at random from each inspection lot and subjected to the applicable tests specified in 4.4. The lots shall be rejected if the sample average fails to meet average requirements, or if more than one sample unit fails to meet sample unit requirements.

4.4 Tests.

4.4.1 Conditioning of test samples. With the exception of item III, test samples shall be conditioned for at least 12 hours in a controlled atmoshpere of 70°F. and 65 percent relative humidity.

4.4.2 Swivel action, item III. The handle shall be securely attached to the brushhead. Place the brushhead flat on the floor and hold the handle in a normal sweeping position. Rotate the brushhead while in a flat position on the floor by twisting the handle and continue twisting the handle until the brushhead has made two complete turns. While still holding the handle lift the brushhead at least 12 inches off the floor and check for compliance with 3.1.3.1.

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4.4.3 Degree of impregnation. The sample items II and IV shall be air-dried for 24 hours or until they feel dry to the touch. A section weighing not less than 30 grams shall be cut from each unit to be tested. In the case of mopheads only the string portion or fringe yarn will be used. The sample shall be placed in a Soxhlet extractor and extracted for 3 hours with toluene. After extraction, the extractables shall be separated from the toluene by applying heat to the mixture by use of a water bath. After separation has been completed, the extractables shall be weighed. The extracted portion of the item shall be air-dried and weighed. The percentage of impregnation shall then be calculated as follows:

$$\text{Percent impregnation} = \frac{\text{Weight of extract}}{\text{Weight of extracted item}} \times 100$$

4.4.4 Oil residue. A specimen consisting of a strip of treated cover cloth and mophead boot material 2-1/2 inches wide and of adequate length shall have a weight of three (3) kilograms placed on its center approximately 2 inches from the end of the specimen that is placed on a black colored nonabsorbent plastic surface. The mophead specimen shall have at least one row of yarn stitched or tufted thereon and shall be placed yarn side down on the plastic surface at the rate of 24 inches per minute. Three specimens each (cover and mophead) shall be tested and if any one specimen yields a deposition of oil on the plastic surface the lot shall be rejected.

4.4.5 Dirt pick-up efficiency.

4.4.5.1 Apparatus. The apparatus consists of a one-quart size mason jar (Owens-Illinois, C2933 Standard round wide mouth jar or equivalent), and a drive motor which permits rotation at 30 RPM (Von Weise Gearmotor, Model VW-8, 115V, 30 RPM, or equivalent), and an interval timer (Dimco-Gray Co., Gray-Lab Timer, Model 171, or equivalent), assembled as shown in figures 1 and 2.

4.4.5.1.1 Assembly of components. The glass tumbler-jar shall be 3-1/2, ± 1/2, inches in outside diameter and 6-1/2, ± 1/2, inches in length and shall be attached horizontally to the shaft of the motor so as to rotate horizontally with its axis. The screw-cap of the jar shall be firmly attached to a circular shaft wheel approximately 4 inches in diameter and 1/2 inch thick. The shaft wheel shall be held firmly on the motor shaft by means of a woodruff key and set screw. This arrangement permits easy attachment and removal of the tumbler-jar from the wheel-screwcap simply by the screwcap principle. Two glass rod baffles shall be securely attached to the inside length of the jar directly opposite each other. The baffles shall be approximately 1/4 inch in diameter. The drive motor shall be mounted on a base to form a stationary or portable support for its operation. See figure 2 for additional details.

4.4.5.2 Synthetic Dirt. A "synthetic dirt" shall be made to the following formula:

Ingredient	Percentages
Walnut sawdust: made with a rotary sanding machine using No. 2 paper on the open grain flat surface of walnut wood	20
Free carbon:	
Lampblack	15
Charcoal, cocoanut, 6-14 mesh	15
Silica:	
80-120 mesh	10
140 mesh	10
240 mesh	10
Pumice powder, N. F.	10
Fuller's Earth, powder, technical	10
	100

4.4.5.3 Preparation of specimen mophead (item II) and brushhead cover (item IV).

4.4.5.3.1 Mophead section. Ten strands of treated fringe yarn shall be cut from the sample mophead and air-dried until a constant weight is obtained. The total treated and air-dried weight of the test specimen shall equal 4 grams, ± 1 gram. The yarns shall be bound together near one end with a suitable thread.

4.4.5.3.2 Brushhead cover section. A specimen consisting of a section 2 inches by 8 inches shall be cut from the sample cover, and then cut again into three unequal parts of any dimensions but not less than 1 inch nor more than 4 inches in length. Record total weight of the clean, treated specimen.

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4.4.5.4 Procedure. The weighed test specimen shall be placed in the tumbler-jar containing 10 grams of the synthetic dirt mixture prepared according to 4.4.5.2, and five rubber stoppers, size 2.

(Note - In regard to the synthetic dirt, provide sufficient quantity for a set of three samples. Carefully divide into uniform 10-gram units. Discard after each test.)

The jar is screwed onto the screwcap of the shaft wheel and the motor is switched on for a period of 5 minutes to permit thorough tumbling and mixing of the contents. The test sample is removed from the jar and weighed. The increase in weight in grams shall represent the weight of dirt pick-up.

4.4.5.5 Computation of dirt pick-up efficiency. The dust-dirt pick-up and retention efficiency shall be calculated as follows:

$$\text{Dirt pick-up, percent} = \frac{F - O}{O} \times 100$$

where: F = Final soiled weight of specimen (see 4.4.5.4).

O = Original weight of clean, treated specimen (see 4.4.5.3).

The percent of dirt pick-up shall be the average of the results obtained from the specimens tested and shall be reported to the nearest 0.1 percent.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. Unless otherwise specified in the contract or order, packaging, packing, and marking shall be in accordance with standard commercial practice.

6. NOTES

6.1 Intended use. This specification is intended to define quality and performance of dust mops and accessories received from commercial rental services.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Items, types, classes and sizes required (see 1.2.1).
(c) Data; inspection responsibility if other than as specified herein (see 4.1).
(d) Whether a laboratory report is not required (see 3.3.1).
(e) Whether directions for use of items are not required (see 3.10).

6.3 Illustrations. Figures 1 and 2 are for illustration purposes and are not intended to preclude other suitable testing equipment.

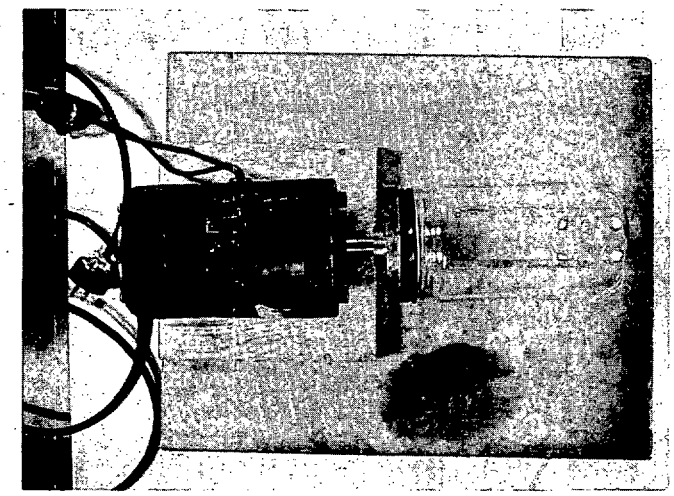
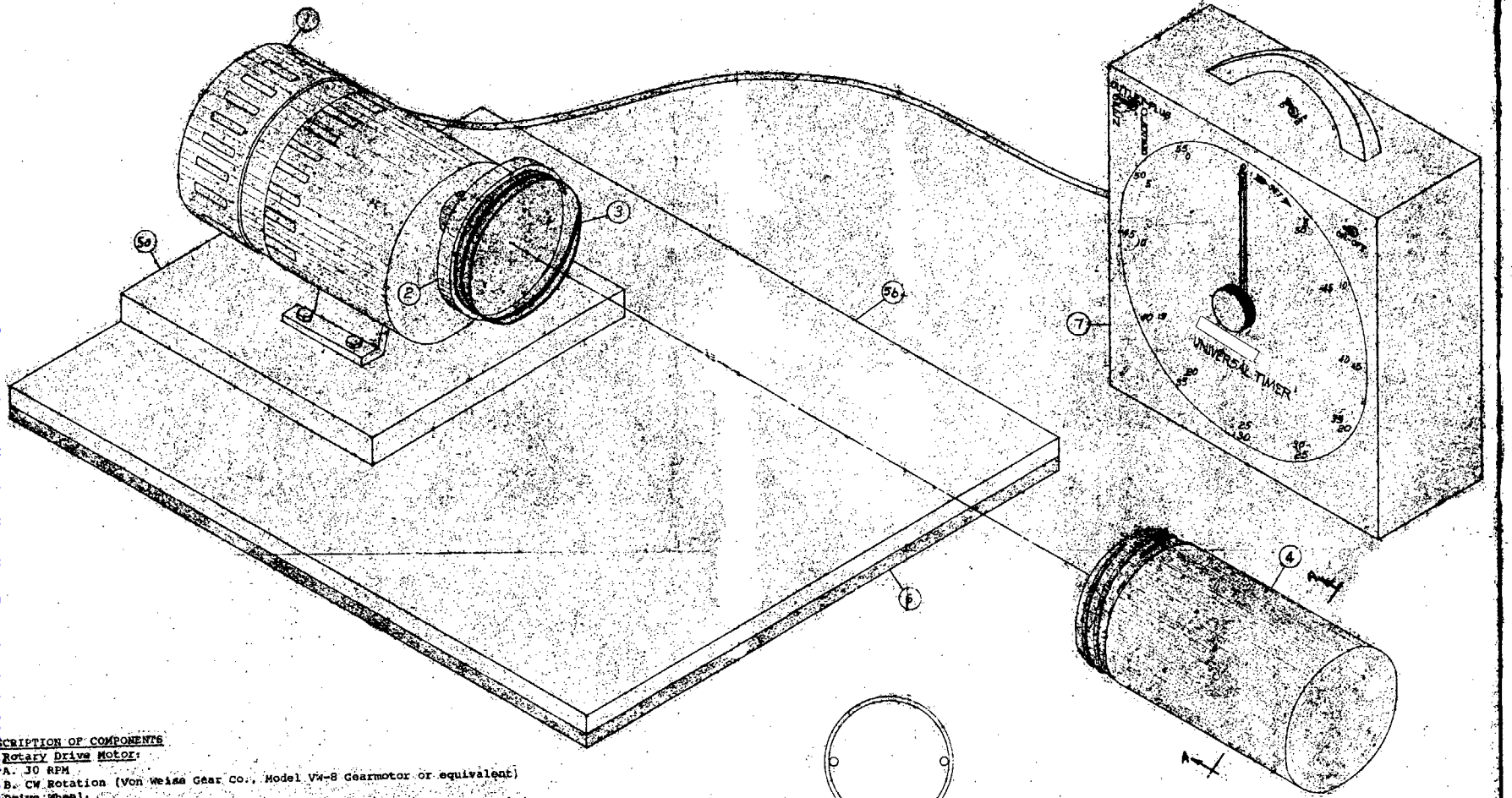


FIGURE 1.

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DESCRIPTION OF COMPONENTS

1. Rotary Drive Motor:
   A. 30 RPM
   B. CW Rotation (Von Weise Gear Co., Model VW-8 Gearmotor or equivalent)
2. Drive Wheel:
   A. Centered and rigidly mounted to drive shaft of Rotary Drive Motor 1
   B. Dimensions and materials as required to insure stable rotation of Jar 4
3. Fixed Drive Cap:
   A. Lid to fit Soiling Jar 4
   B. Lid centered and rigidly fastened to Drive Wheel 2
4. Soiling Jar:
   A. Quart size, glass, Mason jar (Owens-Illinois, C-2933 standard wide mouth jar or equivalent)
   B. Removable
   C. Fitted with two, 1/4 inch (6mm) diameter glass rods equal in length to and permanently adhered to the longitudinally straight inside jar wall at 180° apart and parallel to the axis of operational rotation
5. Base Plate:
   A. Plywood or similar rigid material
   B. Dimensions as required to insure stable rotation during normal operation
6. Dampner Pad:
   A. Vibration damping material, e.g. carpet, foam rubber etc.
   B. Dimensions as required to insure stable rotation during normal operation
7. Interval Timer:
   A. Interval timer of 1-10 minute range in 1 second increments
   B. Integral time switch for shutting off current to Rotary Drive Motor 1 at end of selected time period
   C. Accuracy of 1/50 of 1 percent at maximum interval setting (Dimco-Gray Co., Gray-Lab Timer Model 171 or equivalent)

FIGURE 2.

SOILING TESTER

SCALE

SIZE

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